# MAT 221 - CALCULUS I
# Lesson 19: - Centre of Mass

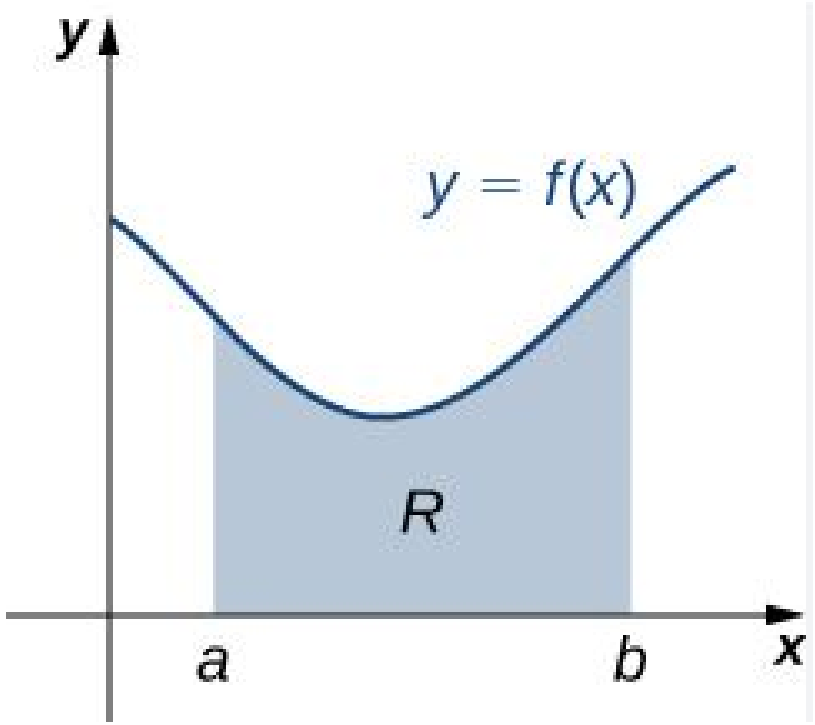
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# Center of Mass or Centroid

- Here we are interested in finding the **center of mass or centroid** of a thin plane with uniform density $\rho$.
- The center of mass or centroid of a region is is the point in which the region will perfectly balanced horizontally if suspended from that point.




- The mass of $R$ is,

$$M = \rho(\text{Area of plate}) = \rho \int_a^b (f(x))\, dx. \tag{1}$$

# Moments

- The moments measure the tendency of the region $R$ to rotate about the $x$ and $y-$axis respectively.
  The moments are given by

$$M_x = \rho \int_a^b \frac{1}{2}(f(x))^2\,dx \tag{2}$$

$$M_y = \rho \int_a^b xf(x)\,dx \tag{3}$$

The coordinates of the center of mass, $(\bar{X}, \bar{Y})$, are then,

$$\bar{X} = \frac{M_y}{M} = \frac{1}{A}\int_a^b x(f(x))\,dx \tag{4}$$

$$\bar{Y} = \frac{M_x}{M} = \frac{1}{2A}\int_a^b ([f(x)]^2)\,dx \tag{5}$$

where

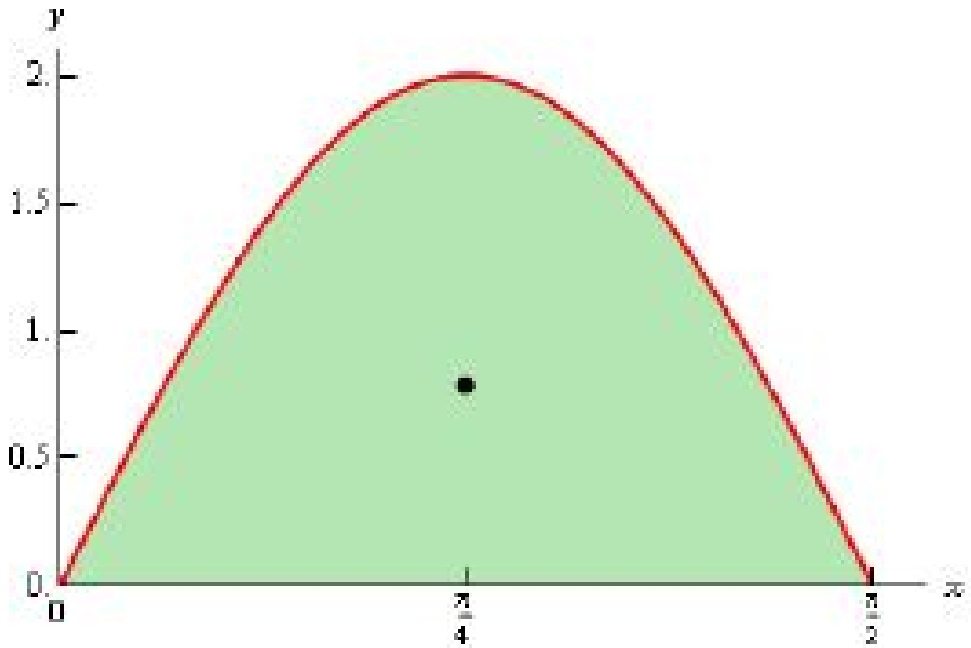
$$A = \int_a^b f(x)\,dx$$

# Example 1

Determine the center of mass for the region bounded by $y = 2\sin(2x)$, $y = 0$ in the interval $\left[0, \frac{\pi}{2}\right]$.

**Solution**

Here is a sketch




The area of the region is

$$A = \int_0^{\frac{\pi}{2}} 2\sin 2x\, dx = -\cos 2x\Big|_0^{\frac{\pi}{2}} = 2$$

Now the moments are

$$
\begin{aligned}
M_x &= \int_0^{\frac{\pi}{2}} 2\sin^2 2x\, dx \\
&= \int_0^{\frac{\pi}{2}} (1-\cos 4x)\, dx \\
&= \frac{\pi}{2} \\
M_y &= \int_0^{\frac{\pi}{2}} 2x\sin 2x\, dx, \qquad \text{Use integration by parts} \\
&= \frac{\pi}{2}
\end{aligned}
$$

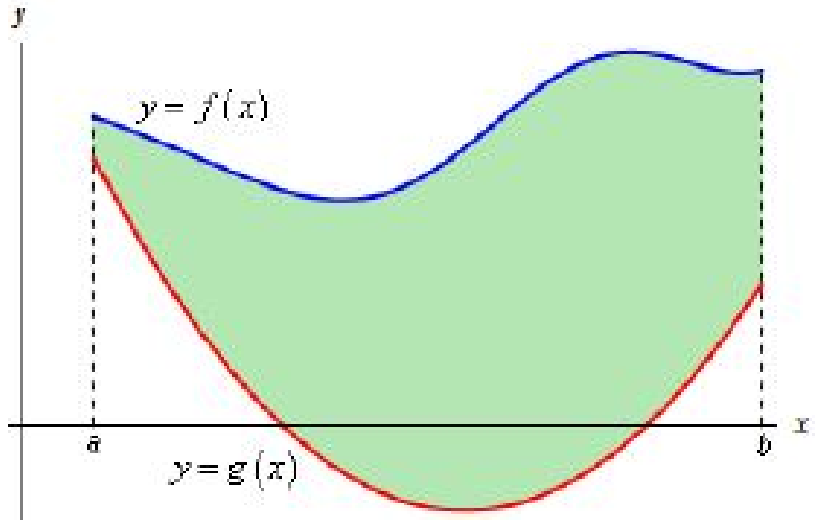
The coordinates of the center of mass are then,

$$\bar{X} = \frac{\frac{\pi}{2}}{2} = \frac{\pi}{4} \quad \text{and} \quad \bar{Y} = \frac{\frac{\pi}{2}}{2} = \frac{\pi}{4}$$

The center of mass is the $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$

# Centroid of A Region by two Curves

- Suppose that the plate is the region bounded by the two curves $f(x)$ and $g(x)$ on the interval $[a, b]$. So, we want to find the center of mass of the region below.




- We'll first need the mass of this plate. The mass is,

$$M = \rho(\text{Area of plate}) = \rho \int_a^b (f(x) - g(x))\, dx. \tag{6}$$

# Moments

- The moments measure the tendency of the region to rotate about the $x$ and $y-$axis respectively.
  The moments are given by

$$M_x = \rho \int_a^b \frac{1}{2}([f(x)]^2 - [g(x)]^2)\, dx \tag{7}$$

$$M_y = \rho \int_a^b x(f(x) - g(x))\, dx \tag{8}$$

The coordinates of the center of mass, $(\bar{X}, \bar{Y})$, are then,

$$\bar{X} = \frac{M_y}{M} = \frac{1}{A} \int_a^b x(f(x) - g(x))\, dx \tag{9}$$

$$\bar{Y} = \frac{M_x}{M} = \frac{1}{2A} \int_a^b ([f(x)]^2 - [g(x)]^2)\, dx \tag{10}$$

where

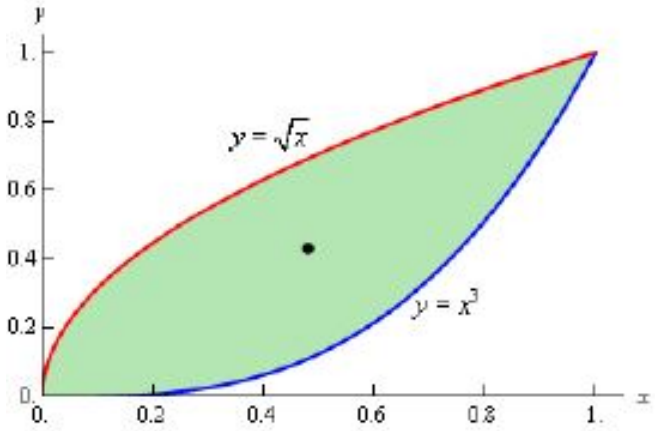
$$A = \int_a^b (f(x) - g(x))\, dx$$

# Example 2

Determine the center of mass for the region bounded by $yx^3$ and $y = \sqrt{x}$.

**Solution**

The two curves intersect at $x = 0$ and $x = 1$ and here is a sketch.




$$A = \int_0^1 (\sqrt{x} - x^3)\,dx = \left(\frac{2}{3}x^{\frac{3}{2}} - \frac{1}{4}x^4\right)_0^1 = \frac{5}{12}, \quad M_x = \int_0^1 \frac{1}{2}(x - x^6)\,dx = \frac{5}{28},$$

$$M_y = \int_0^1 x(\sqrt{x} - x^3)\,dx = \int_0^1 (x^{\frac{3}{2}} - x^4)\,dx = \frac{1}{5},$$

$$\bar{X} = \frac{\frac{1}{5}}{\frac{5}{12}} = \frac{12}{25} \qquad \text{and} \qquad \bar{Y} = \frac{\frac{5}{28}}{\frac{5}{12}} = \frac{3}{7}$$

Center of mass is $\left(\frac{12}{25}, \frac{3}{7}\right)$.